

Callable Statement

--> to call stored procedure
--> Returns Multiple values
--> Create in database
--> IN-input parameter OUT- Returned value

STEPS:

1. Open XAMPP & phpMyAdmin
2. click on database
3. click on Routines
4. Create New Routine
5. Select Procedure
6. write basic syntax

```
CallableStatement cst = con.prepareCall("{call funName()}")
```

Stored Procedure:

```
BEGIN
    SELECT....
    FROM....
    WHERE...
END
```

```
String sql = "{call Procedure(?,?)}";
Callable Statement cst = con.prepareCall(sql);
cst.executeUpdate();
```

```
*/
```

```
import java.sql.*;
import java.util.Scanner;
```

```
public class CallableStatement1 {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        String dburl = "jdbc:mysql://localhost:3306/lju";
        String dbuser = "root";
        String dbpass = "";
        String drivename = "com.mysql.jdbc.Driver";
```

```

// Step 1: Arrange and Load Drivers:

Class.forName(drivername);

// Step 2: Establish Connection:

Connection con = DriverManager.getConnection(dburl, dbuser, dbpass);

// Check is Connection established?

if (con != null)
{
    System.out.println("Connection Sucessful");
}

else
{
    System.out.println("Connection Failed");
}

String sql = "{call getFacultyData()}";
CallableStatement cst = con.prepareCall(sql);
ResultSet rs= cst.executeQuery();

while(rs.next())
{
    System.out.println("\n");
    System.out.println("Faculty Id= "+rs.getInt(1));
    System.out.println("Faculty Name= "+rs.getString(2));
    System.out.println("Faculty salary= "+rs.getFloat(3));
    System.out.println("Faculty Number= "+rs.getString(4));
    System.out.println("\n");
}
}
}

```

getFacultyData

Edit

Details

Routine name

getFacultyData

Type

PROCEDURE

Parameters

Direction	Name	Type	Length/Values	Options
Add parameter				
1	BEGIN			
2	SELECT *			
3	FROM faculty;			
4	END			

Definition

Go

Close

```
// IN PARAMETER
```

```
import java.sql.*;
import java.util.Scanner;
```

```
public class CallableStatement2 {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        String dburl = "jdbc:mysql://localhost:3306/lju";
        String dbuser = "root";
        String dbpass = "";
        String drivename = "com.mysql.jdbc.Driver";

        // Step 1: Arrange and Load Drivers:

        Class.forName(drivename);

        // Step 2: Establish Connection:

        Connection con = DriverManager.getConnection(dburl, dbuser, dbpass);

        // Check is Connection established?

        if (con != null)
        {
            System.out.println("Connection Sucessful");
        }

        else
        {
            System.out.println("Connection Failed");
        }
    }
}
```

```
// IN PARAMETER CALL
```

```
System.out.print("Enter name of faculty which details is given: ");
String s = sc.next();
```

```
String sql = "{call getFac(?)}";
CallableStatement cst = con.prepareCall(sql);
cst.setString(1, s);
ResultSet rs= cst.executeQuery();
```

```

while(rs.next())
{
    System.out.println("\n");
    System.out.println("Faculty Id= "+rs.getInt(1));
    System.out.println("Faculty Name= "+rs.getString(2));
    System.out.println("Faculty salary= "+rs.getFloat(3));
    System.out.println("Faculty Number= "+rs.getString(4));
    System.out.println("\n");
}
}
}

```

getFac

Edit

Details

Routine name

Type

	Direction	Name	Type	Length/Values	Options
Parameters	IN	FAC_NAME	VARCHAR	50	Charset Drop

Add parameter

```

1 BEGIN
2     SELECT *
3     FROM faculty
4     WHERE facName=FAC_NAME;
5 END

```

Definition

Go Close

//OUT PARAMETER

/*

IN: FAC_ID -INT

OUT: FAC_NAME -VARCHAR

getFacByID()

*/

import java.sql.*;

import java.util.Scanner;

public class CallableStatement3 {

public static void main(String[] args) throws Exception {
 Scanner sc = new Scanner(System.in);

String dburl = "jdbc:mysql://localhost:3306/lju";

String dbuser = "root";

String dbpass = "";

String drivename = "com.mysql.jdbc.Driver";

// Step 1: Arrange and Load Drivers:

Class.forName(drivename);

// Step 2: Establish Connection:

Connection con = DriverManager.getConnection(dburl, dbuser, dbpass);

// Check is Connection established?

if (con != null)

{

System.out.println("Connection Sucessful");

}

else

{

System.out.println("Connection Failed");

}

```

// IN-OUT PARAMETER CALL

System.out.print("Enter id of faculty which name is required: ");
int i = sc.nextInt();

String sql = "{call getFacByID(?,?)}";
CallableStatement cst = con.prepareCall(sql);
cst.setInt(1, i);
cst.execute();
String s = cst.getString(2);
System.out.println("Faculty Name: "+s);
}
}

```

getFacByID

Edit

Details

Routine name

getFacultyId

Type

PROCEDURE

Parameters

	Direction	Name	Type	Length/Values	Options	
↑	IN	FAC_ID	INT	10		Drop
↓	OUT	FAC_NAME	VARCHAR	50	Charset	Drop

Add parameter

```

1 BEGIN
2 SELECT facName INTO FAC_NAME
3 FROM faculty
4 WHERE facId =FAC_ID;
5 END

```

Go

```

import java.sql.*;
import java.util.Scanner;

public class CallableStatement4 {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        String dburl = "jdbc:mysql://localhost:3306/lju";
        String dbuser = "root";
        String dbpass = "";
        String drivename = "com.mysql.jdbc.Driver";

        // Step 1: Arrange and Load Drivers:

        Class.forName(drivename);

        // Step 2: Establish Connection:

        Connection con = DriverManager.getConnection(dburl, dbuser, dbpass);

        // Check is Connection established?

        if (con != null)
        {
            System.out.println("Connection Sucessful");
        }

        else
        {
            System.out.println("Connection Failed");
        }

        // IN-OUT PARAMETER CALL

        System.out.print("Enter name of faculty which you want to insert: ");
        String nName = sc.next();
        System.out.print("Enter Salary of faculty which you want to insert: ");
        float nSalary = sc.nextFloat();
        System.out.print("Enter Number of faculty which you want to insert: ");
        String nNumber = sc.next();
        System.out.print("Enter id of faculty for which you want to update faculty
name: ");
        int upID= sc.nextInt();
    }
}

```

```
System.out.print("Enter name of faculty which you want to update: ");
String upName= sc.next();
```

```
String sql = "{call manipulateFacData(?,?,?,?,?)}";
CallableStatement cst = con.prepareCall(sql);
cst.setString(1,nName );
cst.setFloat(2,nSalary );
cst.setString(3,nNumber );
cst.setInt(4, upID);
cst.setString(5, upName);
cst.executeUpdate();
System.out.println("Executed Successfully");

}
}
```

manipulateFacData

Edit

Details

Routine name
manipulateFacData

Type
PROCEDURE

	Direction	Name	Type	Length/Values	Options
Parameters	IN	In_Name	VARCHAR	50	Charset
	IN	In_Salary	FLOAT	50	
	IN	In_Number	VARCHAR	50	Charset
	IN	Update_Name	VARCHAR	50	Charset
	IN	Update_Id	INT	10	

Add parameter

```

1 BEGIN
2 INSERT INTO faculty(facName,salary,number) values(In_Name,In_Salary,In_Number);
3 UPDATE faculty set facName=Update_Name WHERE facId=Update_Id;
4 END

```

Definition

Go Close

```

import java.sql.*;
import java.util.Scanner;

public class CallableStatement5 {
    public static void main(String[] args) throws Exception {
        Scanner sc = new Scanner(System.in);

        String dburl = "jdbc:mysql://localhost:3306/lju";
        String dbuser = "root";
        String dbpass = "";
        String drivename = "com.mysql.jdbc.Driver";

        // Step 1: Arrange and Load Drivers:

        Class.forName(drivename);

        // Step 2: Establish Connection:

        Connection con = DriverManager.getConnection(dburl, dbuser, dbpass);

        // Check is Connection established?

        if (con != null)
        {
            System.out.println("Connection Sucessful");
        }

        else
        {
            System.out.println("Connection Failed");
        }

        // IN-OUT PARAMETER CALL

        System.out.print("Enter name of faculty which you want to update: ");
        String upName = sc.next();
        System.out.print("Enter Salary of faculty which you want to update: ");
        float upSalary = sc.nextFloat();
        System.out.print("Enter id of faculty in which you want to update name & salary: ");
        String upID = sc.next();
        System.out.print("Enter id of faculty which name & salary is required: ");
        int selID= sc.nextInt();
    }
}

```

```

String sql = "{call updateSelectFacData(?,?,?,?)}";
CallableStatement cst = con.prepareCall(sql);
cst.setString(1,upName );
cst.setFloat(2,upSalary );
cst.setString(3,upID );
cst.setInt(6, selID);
cst.executeUpdate();
System.out.println("Executed Successfully");
System.out.println("Faculty Name: "+cst.getString(4));
System.out.println("Faculty Salary: "+cst.getFloat(5));

}
}

```

updateSelectFacData

Edit

Details

Routine name

Type
PROCEDURE

Parameters

Direction	Name	Type	Length/Values	Options
IN	Up_Name	VARCHAR	50	Charset
IN	Up_Salary	FLOAT	50	
IN	Up_Id	INT	10	
OUT	select_Name	VARCHAR	50	Charset
OUT	select_Salary	FLOAT	50	
IN	select_Id	INT	10	

Add parameter

```

1 BEGIN
2 UPDATE faculty set facName=Up_Name,salary=Up_Salary WHERE facId=Up_Id;
3 SELECT facName,salary INTO select_Name,select_Salary From faculty where facId=select_Id;
4 END

```

Definition

Go Close